# ASR-12 Primary Surveillance Radar

Representative parameter datasheet for MATLAB agentic-workflow exercises

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Classification: Training Use Only

**IMPORTANT - SIMULATION DATA ONLY.**

This is a fictional, representative training artifact for classroom radar modeling. Values, site name, and configuration are not manufacturer, FAA, airport, or operational specifications and must not be used for navigation, procurement, safety, or real-world system configuration.

## Terminal-area primary surveillance

Rotating S-band radar configuration designed as a coherent parameter set for a classroom air-traffic-control scenario and a first-pass MATLAB radar simulation.

| 240 | 60 | 2.8 |
|---|---|---|
| NMI INSTRUMENTED RANGE | RPM SCAN RATE | GHZ NOMINAL FREQUENCY |

### Configuration at a glance

| | |
|---|---|
| Radar role | **Primary surveillance** |
| Scenario ID | **KMTN-TRN-01** |
| Coverage mode | **Terminal / 360 deg** |
| Propagation model | **Free-space baseline** |
| Target class | **Fixed-wing aircraft** |
| Configuration | **Baseline terminal coverage** |

## 1. System identity and operating envelope

| Parameter | Representative value |
|---|---|
| System designation | ASR-12 (training representative) |
| Radar type | Coherent pulsed primary surveillance radar |
| Frequency band | S-band |
| Instrumented range | 0-240 nmi (0-444 km) |
| Azimuth coverage | 0-360 deg |
| Elevation coverage | 20 deg instantaneous vertical field of view |
| Environmental baseline | Standard atmosphere; clear air |

## 2. Primary model input set

| Parameter | Representative value |
|---|---|
| Carrier frequency, fc | 2.80 GHz |
| Peak transmit power, Pt | 1.10 MW |
| Pulse repetition frequency, PRF | 325 Hz |
| Transmit pulse width, tau | 2.15 us |
| Antenna peak gain, G | 37.21 dBi |
| Azimuth 3 dB beamwidth | 1.4 deg |
| Elevation 3 dB beamwidth | 4.5 deg |
| Antenna scan rate | 60 rpm |
| Receiver noise figure | 4.0 dB |
| System losses | 6.0 dB |
| Reference target RCS | 10 m2 |

# ASR-12 Modeling Notes

## 3. Detection and measurement assumptions

| Parameter | Representative value | Use in exercise |
|---|---|---|
| Probability of detection, Pd | 0.90 at reference condition | Reference only: 10 m2 target at 240 nmi, boresight, clear air. |
| Probability of false alarm, Pfa | 1.0e-6 per resolution cell | Set detector threshold or use as a stated evaluation constraint. |
| Range measurement accuracy | +/-116 m (exercise assumption) | Add zero-mean Gaussian range error for track reports. |
| Azimuth measurement accuracy | +/-0.176 deg (exercise assumption) | Add zero-mean Gaussian bearing error for track reports. |
| Range resolution (nominal) | 323 m | Use c*tau/2; no pulse compression in baseline configuration. |
| Unambiguous range (baseline) | 461 km (249 nmi) | At 325 Hz; exceeds the 240 nmi reporting gate. |
| Update interval | 1.0 s | One measurement opportunity per antenna revolution. |

## 4. Classroom air-traffic-control scenario

### Radar site

Fictional KMTN Training Terminal, reference position [0, 0, 0] m in local ENU coordinates. Antenna phase center height: 10 m above ground level.

### Targets at scenario start

| Track | Position [E, N, U] m | Velocity [E, N, U] m/s | RCS |
|---|---|---|---|
| TRN101 | [18,000, 65,000, 2,800] | [-115, -40, 0] | 10 m2 |
| TRN205 | [-32,000, 48,000, 4,100] | [95, -65, -2] | 6 m2 |
| TRN317 | [7,000, 26,000, 1,700] | [45, 130, 3] | 3 m2 |

### Exercise scope

- Import this document and extract a structured parameter set.
- Build a rotating monostatic radar model.
- Simulate range/azimuth detections for the three targets.
- Compare ideal versus noisy measurement reports over 60 s.

### Deliberately excluded from baseline

secondary surveillance · IFF · terrain masking · weather clutter · multipath · target fluctuation

## 5. Data-quality notes

| Rule | Expected behavior |
|---|---|
| Operating units | Values are stated in the customary units used in this reference document. |
| Derived quantities | Range resolution, unambiguous range, and scan period are nominal values based on the listed baseline configuration. |
| Configuration control | Changes to the listed values require a corresponding revision to this training reference. |
| Document status | Fictional training reference; not an operational, procurement, or maintenance specification. |

The 325 Hz baseline waveform gives approximately 461 km (249 nmi) unambiguous range, exceeding the 240 nmi instrumented-range requirement.